

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                               |
|----------------------|-------------------------------|
| Product Name         | <u>Boron oxide</u>            |
| CAS No               | 1303-86-2                     |
| Synonyms             | Boron trioxide                |
| Molecular Formula    | B <sub>2</sub> O <sub>3</sub> |
| Molecular Weight     | 69.61                         |
| Recommended Use      | Laboratory chemicals.         |
| Uses advised against | No Information available      |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | 206300000; 206300010; 206300050; 206300250                                                  |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

HSNO Approval Number     HSR002503

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                   |             |
|-----------------------------------|-------------|
| Skin Corrosion/Irritation         | Category 2  |
| Serious Eye Damage/Eye Irritation | Category 2  |
| Reproductive Toxicity             | Category 1B |

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



Signal Word

Danger

**Hazard Statements**

H360 - May damage fertility or the unborn child

H315 - Causes skin irritation

H319 - Causes serious eye irritation

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P308 + P313 - IF exposed or concerned: Get medical advice/attention

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

Other hazards which do not result in classification

## Section 3 - Composition and Information on Ingredients

| Component   | CAS No    | Weight % |
|-------------|-----------|----------|
| Boron oxide | 1303-86-2 | >95      |

## Section 4 - First Aid Measures

**Description of first aid measures****New Zealand Emergency Tel.**CHEMTREC®  
09 980 6780 or +64 9 980 6780**Inhalation**

Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**First Aid Facilities**

Eyewash, safety shower and washroom.

**Most important symptoms and effects**

No information available.

**Notes to Physician**

Treat symptomatically.

## **Section 5 - Fire Fighting Measures**

**Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Non-combustible.

**Hazardous Combustion Products**

Oxides of boron.

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **Section 6 - Accidental Release Measures**

**Personal Precautions, Protective Equipment and Emergency Procedures****Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Use only under a chemical fume hood. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Incompatible Materials**

Strong oxidizing agents. Acids. Fluorine. Strong reducing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component   | New Zealand WEL           | Australia                 | ACGIH TLV                 | The United Kingdom                                                  |
|-------------|---------------------------|---------------------------|---------------------------|---------------------------------------------------------------------|
| Boron oxide | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> | STEL: 20 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

#### Recommended Filter type: Recommended half mask:-

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)  
Particle filtering: EN149:2001 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid                    |                                          |
| <b>Appearance</b>                              | White                    |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | 4                        | 10 g/L (25°C)                            |
| <b>Melting Point/Range</b>                     | 450 °C / 842 °F          |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 1860 °C / 3380 °F        | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Water Solubility</b>                        | 36 g/L (25°C)            |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 2.460                    |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Particle characteristics</b>                | No data available        |                                          |
| <b>Other information</b>                       |                          |                                          |
| <b>Molecular Formula</b>                       | B2 O3                    |                                          |
| <b>Molecular Weight</b>                        | 69.61                    |                                          |
| <b>Evaporation Rate</b>                        | Not applicable - Solid   |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                |
| <b>Stability</b>                        | Hygroscopic.                                                                              |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                  |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                  |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                             |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Avoid dust formation, Exposure to moist air or water. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Acids, Fluorine, Strong reducing agents.                         |
| <b>Hazardous Decomposition Products</b> | Oxides of boron.                                                                          |

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

|            |                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------|
| Inhalation | May cause irritation of respiratory tract. May be harmful if inhaled.                                        |
| Eyes       | May cause irritation.                                                                                        |
| Skin       | May cause irritation. May be harmful in contact with skin.                                                   |
| Ingestion  | May be harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |

### Numerical measures of toxicity

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Based on available data, the classification criteria are not met |
| Dermal              | No data available                                                |
| Inhalation          | No data available                                                |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

|                                        |                   |
|----------------------------------------|-------------------|
| (d) respiratory or skin sensitization; |                   |
| Respiratory                            | No data available |
| Skin                                   | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

|                            |                                    |
|----------------------------|------------------------------------|
| (g) reproductive toxicity; | Category 1B                        |
| Reproductive Effects       | May impair fertility               |
| Developmental Effects      | May cause harm to the unborn child |

(h) STOT-single exposure; No data available

|                             |                           |
|-----------------------------|---------------------------|
| (i) STOT-repeated exposure; | No data available         |
| Target Organs               | No information available. |

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed  
No information available.

## Section 12 - Ecological Information

**Ecotoxicity****Aquatic ecotoxicity** Do not empty into drains. .

| Component   | Freshwater Fish                           | Water Flea                                   | Freshwater Algae | Microtox |
|-------------|-------------------------------------------|----------------------------------------------|------------------|----------|
| Boron oxide | LC50: 570 mg/L/72h<br>(Carassius auratus) | EC50: 370 - 490 mg/L,<br>48h (Daphnia magna) |                  |          |

**Terrestrial ecotoxicity**

| Component   | Earthworm | Avian                                                                                                                                         | Honeybees |
|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Boron oxide |           | Dietary toxicity: LC50 > 3160<br>ppm (Anas platyrhynchos, 5<br>Days)<br>Dietary toxicity: LC50 > 3160<br>ppm (Colinus virginianus, 5<br>Days) |           |

**Persistence and Degradability** Not readily biodegradable**Persistence** Soluble in water, Persistence is unlikely, based on information available.**Degradability** Not relevant for inorganic substances.**Bioaccumulative Potential** Bioaccumulation is unlikely**Mobility** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils**Other adverse effects**

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods**

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## Section 14 - Transport Information

**NZS 5433:2020** Not regulated

|                                                                                 |                                                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IATA</b>                                                                     | Not regulated                                                                                                           |
| <b>IMDG/IMO</b>                                                                 | Not regulated                                                                                                           |
| <b>Environmental hazards</b>                                                    | No hazards identified                                                                                                   |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                                          |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| <b>Additional information</b>                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR002503 |
|-----------------------------|-----------|

#### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

### International Regulations

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| <b>Ozone Depletion Potential</b>    | This product does not contain any known or suspected substance |
| <b>Persistent Organic Pollutant</b> | This product does not contain any known or suspected substance |
| <b>Rotterdam Convention (PIC)</b>   | Not applicable                                                 |

#### **Authorisation/Restrictions according to EU REACH**

| Component   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Boron oxide | -                                                                   | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - Toxic for reproduction (Article 57 c)                                           |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.



<https://echa.europa.eu/authorisation-list>  
<https://echa.europa.eu/substances-restricted-under-reach>  
<https://echa.europa.eu/candidate-list-table>

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component   | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Boron oxide | 1303-86-2 | X     | X    | 215-125-8 | -      | -   | KE-09919 | X     | X    |

| Component   | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Boron oxide | 1303-86-2 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date**

10-Mar-2023

**Revision Summary**

Not applicable

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**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**